Chapter 1, Problem 9P

Problem

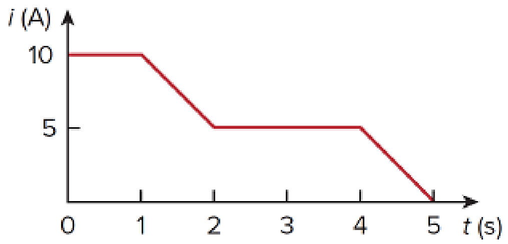
The current through an element is shown in Fig. 1.26. Determine the total charge that passed through the element at:

(a) $t = 1$ s

(b) $t = 3$ s

(c) $t = 5$ s

Figure 1.26:



Step-by-step solution

Step 1 of 3

Refer to Figure 1.26 in the textbook.

(i)

Consider the time, $t = 1\text{ s}$.

Determine the total charge that passed through the element.

$$\begin{aligned} q(t) &= \int_0^1 10dt \\ &= [10t]_0^1 \\ &= 10(1-0) \\ &= 10\text{ C} \end{aligned}$$

Thus, the total charge that passed through the element is $\boxed{10\text{ C}}$.

Step 2 of 3

(ii)

Consider the time, $t = 3\text{ s}$.

Determine the line equation from points $(1,10)$ and $(2,5)$.

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1}$$
$$\frac{5-10}{2-1} = \frac{i(t)-10}{t-1}$$
$$i(t) - 10 = -5(t-1)$$
$$i(t) = -5t + 15$$

Determine the total charge that passed through the element.

$$\begin{aligned} q(t) &= \int_0^3 i(t)dt \\ &= \int_0^1 10dt + \int_1^2 (-5t+15)dt + \int_2^3 5dt \\ &= 10(1-0) + \left[-5\frac{t^2}{2} + 15t\right]_1^2 + 5(3-2) \\ &= 10 + 5 + \left[\left(-\frac{5\times 2^2}{2} + 15\times 2\right) - \left(-\frac{5\times 1^2}{2} + 15\times 1\right)\right] \end{aligned}$$

$$\begin{aligned} q(t) &= 15 + [(-10+30) - (12.5)] \\ &= 15 + 7.5 \\ &= 22.5\text{ C} \end{aligned}$$

Thus, the total charge that passed through the element is $\boxed{22.5\text{ C}}$.

Step 3 of 3

(iii)

Consider the time, $t = 5\text{ s}$.

Determine the line equation from points $(4,5)$ and $(5,0)$.

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1}$$
$$\frac{0-5}{5-4} = \frac{i(t)-5}{t-4}$$
$$i(t) - 5 = -5(t-4)$$
$$i(t) = -5t + 25$$

Determine the total charge that passed through the element.

$$\begin{aligned} q(t) &= \int_0^5 i(t)dt \\ &= \int_0^1 10dt + \int_1^2 (-5t+15)dt + \int_2^4 5dt + \int_4^5 (-5t+25)dt \\ &= 10(1-0) + \left[-5\frac{t^2}{2} + 15t\right]_1^2 + 5(4-2) + \left[\frac{-5}{2}t^2 + 25t\right]_4^5 \\ &= \left\{ \begin{array}{l} 10 + \left[\left(-\frac{5\times 2^2}{2} + 15\times 2\right) - \left(-\frac{5\times 1^2}{2} + 15\times 1\right)\right] + 10 \\ \qquad + \left[\left(-\frac{5\times 5^2}{2} + 25\times 5\right) - \left(-\frac{5\times 4^2}{2} + 25\times 4\right)\right] \end{array} \right\} \end{aligned}$$

$$\begin{aligned} &= 10 + [(-10+30) - (-2.5+15)] + 10 + [(-62.5+125) - (-40+100)] \\ &= 10 + [7.5] + 10 + [2.5] \\ &= 30\text{ C} \end{aligned}$$

Thus, the total charge that passed through the element is $\boxed{30\text{ C}}$.